

Research Note: Does NIFTY 50 Reliably Recover After a Significant One-Day Fall?

1. Hypothesis

A common market heuristic suggests that sharp, sudden market drops are followed by immediate "bounce-back" recoveries as investors buy the dip. This study quantitatively investigates whether the NIFTY 50 index reliably recovers over the following 1, 3, and 5 trading days after a significant one-day fall (defined as a daily close-to-close drop of -2.0% or worse).

2. Methodology

To test this hypothesis without look-ahead bias, a configurable event-driven backtesting engine was constructed:

- Data Source: Daily OHLC data for the NIFTY 50 index (^NSEI) from January 2011 to January 2026.
- Event Criteria: Daily Close-to-Close return $\leq -2.0\%$.
- Execution & Holding: To simulate realistic execution, entry is set at the *Next Day's Open* following the crash event, with exits captured at the Close of Day 1, Day 3, and Day 5.

3. Evidence & Baseline Comparison

Over the 15-year dataset, the engine identified 94 instances of a -2.0% or worse drop.

- Forward Returns (Crash Events):
 - 1-Day Holding: 46.81% Win Rate | 0.10% Mean Return
 - 3-Day Holding: 58.51% Win Rate | 0.23% Mean Return
 - 5-Day Holding: 53.19% Win Rate | 0.12% Mean Return
- Market Baseline (All Days): The 3-Day baseline win rate across all historical trading days is 52.70% (Mean Return: 0.02%). While the crash event shows a slightly elevated 3-day win rate (58.51%), we must test if this is statistically significant.

4. Robustness & Statistical Testing

To verify if the 3-day recovery represents a genuine edge or random noise, we performed Welch's T-Tests against normal market returns:

- -2.0% Threshold Test: Yielded a t-statistic of 0.6468 and a p-value of 0.5193. Because $p > 0.05$, the observed outperformance is statistically insignificant.

- Sensitivity Test (-3.0% Drop): Tightening the threshold to -3.0% yielded 25 events. The win rate increased to 68.00% (Mean Return: 0.76%), but the p-value remained high at 0.4116 due to the severely small sample size.

5. Out-of-Sample Validation

To prevent overfitting, the data was split chronologically:

- In-Sample (2011–2021): 80 events | 57.50% Win Rate | $p = 0.7958$
- Out-of-Sample (2022–2026): 14 events | 64.29% Win Rate | $p = 0.1884$

The pattern failed to achieve statistical significance ($p < 0.05$) in both regimes.

6. Limitations

1. Transaction Costs: Gross returns do not account for brokerage fees, slippage on next-day opens, or STT taxes.
2. Regime Dependence: The study aggregates all market environments without filtering for structural bull vs. bear regimes.
3. Survivorship Bias: NIFTY 50 constituents change periodically, introducing an inherent upward drift.

7. Conclusion

The hypothesis that the NIFTY 50 index reliably recovers following a significant crash is rejected.

Although 3-day forward returns show a minor visual elevation over the baseline, rigorous statistical testing and out-of-sample validation confirm that this variance is indistinguishable from random market noise. Blindly buying the dip based solely on single-day magnitude does not constitute a viable trading edge.